Vertically Integrated Industries

Initial takeaways on minerals, metals, manufacturing and chemistry

1. Tailwinds, key takeaways and macro

This theme covers companies that make physical things in their own plants, such as metal parts, magnets and molecules, and use their own software and AI to make them faster and with fewer skilled people than older competitors. They earn money on what they ship. We call them vertically integrated because one company owns the whole job, from taking the order to shipping the product. The takeaways below cover why demand is rising, what makes these businesses hard to build, and where the money ends up.

1. **Some tailwinds behind US production are lasting and others can reverse. The lasting ones are what we should build a thesis on.**
   - **Lasting:** China's control of mineral refining and magnets, the shortage of skilled factory workers, and the years it takes a big buyer to approve a new supplier. None of these goes away with an election or a budget cycle.
   - **Temporary:** tariffs, the current surge in munitions spending and interest rates. Tariffs are set by proclamation and can change overnight. Munitions money depends on bills Congress has not passed. The Fed raised rates to 3.75-4.00% in September, and spending on new factory construction was down 21.9% year on year by May 2026 (Federal Reserve, ENR).
   - **Implications:** we should favor companies that still work if tariffs fall and the munitions surge slows. That means owning approved capacity that buyers need for a decade, and avoiding businesses whose margin exists only because of a tariff.
2. **China's hold on minerals and magnets is the most lasting limit in the theme.**
   - China leads refining in 19 of 20 strategic minerals, with about 70% average share (IEA). Refining is the step that turns ore into pure metal. China also makes about 94% of the world's rare-earth magnets (Mining.com). These magnets go into drones, missiles, motors and robots.
   - The US is fully import-reliant for 16 critical minerals (USGS). China has already used export controls on these materials, and its current pause on those controls ends on November 10 and November 27, 2026.
   - This is why buyers and the government will pay for a non-Chinese supply even when it costs more. It also means the plants that fix this problem are expensive and slow to build.
3. **Defense demand and government money are real, but much of the money is conditional and slow.**
   - The FY27 budget request asks for $49.5B for 10 key munitions, about 8.5 times the past decade's average. About 82% of that depends on a reconciliation bill Congress has not passed (CNAS). The Navy is putting $900M into plants that make submarine parts (Breaking Defense).
   - The FY26 defense bill lets the Pentagon buy munitions over several years and commit to purchases of up to 10 years, with advance payments. That is new and helpful, though each year still needs Congress to approve the money (FY26 NDAA).
   - Government now acts as lender and investor. The Pentagon's Office of Strategic Capital (OSC), which lends to companies in key supply chains, has made a $620M loan to one magnet maker and a $500M conditional loan to a rare-earth metals maker, and about $20B is requested for FY27 (Mintz). The Department of Energy's loan office has over $289B of lending authority (DOE). Governments worldwide put about $65B into minerals in 2025 (IEA). These loans take a long time to close and come with conditions.
4. **The labor gap is lasting, and it is driving owners of small shops to sell.**
   - US manufacturers need to fill about 3.8M jobs by 2033, and about 1.9M of them may go unfilled (Deloitte).
   - Of 17,156 US machine shops, 83% have fewer than 20 employees (Census). Many owners are near retirement with no one to take over. That creates a supply of certified shops for sale at reasonable prices.

- Fewer skilled people means a plant that needs fewer of them can grow where others cannot. That is the opening for software and automation.

5. **Capex before contracts is the central risk. A factory has to spend its money before it wins the orders.**
   - Capex means capital spending on buildings, machines and certifications. A big defense contractor, often called a prime, won't approve a new supplier or sign a volume contract until the plant exists and is certified. Approval takes about nine months to two years: one prime told us 9-12 months in an expert call, and another said one to two years.
   - The numbers are large. One precision-parts maker committed about $200M to a single Arizona plant (Plant Services), and its CEO says capex runs about two years ahead of revenue (claim). Process plants need about 80-85% use of capacity to make money (FTI Consulting). A hardware company at Series B often has about $10M of revenue and break-even gross margins (Lowercarbon).
   - This hurts venture returns. Equity spent on a building earns building-like returns and dilutes founders before the first big order arrives.
   - The winners shift who pays for the build. Options include government loans and grants, a buyer that agrees to purchase output before the plant is built (the Pentagon agreed to buy all of one magnet plant's output for 10 years, with a price floor), a customer that hosts the factory inside its own plant (one 7-year deal), bank debt once revenue exists, leased buildings, franchising, and buying plants that are already certified. One roll-up founder said they buy shops at about seven times annual earnings (claim).
6. **Buyers pay for speed and spare capacity. Few pay for a lower price, so the margin goes to whoever owns the output and holds the approvals.**
   - "Cost is secondary. Cost is driven usually by schedule," a director at a large defense contractor said in an expert call. An engineer there would pay 20-40% more to avoid a delay. The premium is highest on prototypes, spare parts and urgent orders, and it fades at volume.
   - Owning the plant pays. One listed parts maker earns a 49% gross margin in its own plants and 31% on work it farms out to partner shops (company filings). Approved incumbents earn well too: a large engine-parts unit runs a 37.7% EBITDA margin (earnings before interest, taxes, depreciation and amortization). Aerospace programs can run about 50 years, so an approval keeps paying.
   - Selling software or machines to factories and mines has been hard. One 3D-printer maker fell from a $2.1B value to a $42.5M sale (3D Printing Journal). An AI-run machining startup in Germany went insolvent in August 2026. Buyers cap each supplier at about a quarter to a third of their work, so even a winner grows one share at a time.
7. **AI's edge today is in the work around the machine and in processes that run nonstop. Nobody has shown yet that it makes each unit cheaper as volume grows.**
   - Around the machine means quoting, planning, writing the instructions a machine follows (CAM programming), inspection and paperwork. One startup claims it cut assembly-instruction writing from about 10 hours to 15 minutes (claim). A former employee of one parts maker said its time from order to first part fell about 50% over two years.
   - The machines themselves are underused. In a survey of more than 100 plants, cutting machines ran about 34% of the time against a 70% target. European process plants run at about 74% of capacity against the 80-85% they need (FTI Consulting). Better planning and control software closes that gap.
   - In a plant that runs nonstop, like a chemical or metals line, every extra point of yield (good output per unit of input) is close to pure profit. A head of strategy at a large aircraft maker put net savings from automation, after paying for the machines, at 20-30% at best. Labor is under 10% of the cost of goods in a new automated plant (claim), so the case rests on speed and yield more than headcount.
8. **The key risk is that incumbents automate and buy faster than new companies can grow.**

- The primes decide who is approved and how much work each supplier gets. The largest aerospace parts supplier saw its share of one buyer's machining fall from about 50% to 30%, and it is now automating (expert call).
- Big companies buy instead of build. Miners back startups through their venture arms, one paid $6.7B for a lithium producer, and a defense contractor's venture arm led a round in an AI materials startup. Drug companies are building their own AI labs, one with plans for up to $1B.
- There is still room. The Pentagon relies on more than 200,000 suppliers and has country-of-origin data for fewer than 10% of the lower-tier suppliers on the F-35 (GAO). Most of that base is small, old and short of people, which is where new companies can win.

2. Interesting categories

This section follows the categories in our primer and covers each one where our view is positive or possibly positive. Each ends with the companies the primer names there, so we can decide together which to keep.

A. Raw Materials

China's hold on refining makes this the most lasting tailwind in the theme, but these are also the most capital-heavy businesses. Our primer view is selective: attractive in refining when someone else pays for the plant, and early in exploration.

Stage A: Locating & mining critical minerals

- **Why it's interesting:** Finding a deposit is slow and mostly fails; one explorer puts success at about 0.5% per project (claim), and US mines take about 19 years from discovery to production. AI can narrow where to drill. Data centers and the grid also need far more copper, so restarting old, permitted mines with automation is attractive.
- **How the business works:** Explorers own the claims and drill rigs, prove out a deposit and sell it to a miner, for example for about 5% of the metal in the ground plus a royalty (claim). That costs far less than a mine, but revenue comes in lumps years apart. Miners own the mine and sell metal at market prices, which is fully owned and very capital-heavy: about 5 years to build and 3 to 5 to ramp after permits (claim). Software sold to miners on its own rarely sticks.

Example companies: Earth AI, KoBold Metals, Terra AI, GeologicAI, Ideon, Mariana Minerals, BHP, Rio Tinto, Caterpillar

Stage B: Refining minerals/metals

- **Why it's interesting:** Chinese refineries take about 6 months to commission against 2 to 4 years in the West. Defense and auto buyers need non-Chinese metal and magnets, and the government is lending and signing long purchase deals to get plants built. Tariffs help too: one large steelmaker's margin rose from 14% to 20%.
- **How the business works:** The company buys concentrate or scrap and sells refined oxides, metals or magnets under long contracts. It owns the plant, which costs hundreds of millions of dollars. The best deals push that cost onto others: a $620M government loan to one magnet maker, and for the largest incumbent a 10-year price floor and a promise that all output of its next magnet plant will be bought. Without a floor, a US refiner takes prices set in China, and equity-funded refineries are the capex-before-contracts problem at its worst. Even the largest US producer lost $85.9M on $224.4M of revenue in 2025.

Example companies: Vulcan Elements, Phoenix Tailings, Mariana Minerals, ReElement, Ucore, Energy Fuels, MP Materials, Hertha, Cyclic, Mass Magnetics, Duranium, Boston Metal, Electra, Ceibo, Nucor, Carpenter Technology

B. Manufacturing

Manufacturing is the most commercial part of the theme, and most companies here already have revenue. Our primer view is attractive for fabrication, selective for precision and defense hardware, and low conviction for metal supply.

Metal Supply

- **Why it's interesting:** Tariffs and buy-American rules help US metal, and job shops wait days for a quote from a broker. An online distributor can quote in minutes (claim).
- **How the business works:** The distributor buys metal from mills, holds it in stock, cuts it to size and sells it to shops and primes. It earns the spread plus cutting fees. Plants are cheap, but a lot of cash sits in inventory. Incumbents are gaining share without AI, so this is our lowest-conviction manufacturing category.

Example companies: Reliance, Ryerson

Fabrication

- **Why it's interesting:** Fabrication means cutting, bending and welding metal into frames and enclosures, and assembling parts like wire harnesses, the bundles of wires that carry power and signals through an aircraft or missile. Every kind of hardware company needs this fast. One online fabricator handles about 32,000 orders a month, is growing about 100% a year and says it has been profitable for eight years (claim). One harness maker claims it cut lead time from six months to two weeks (claim).
- **How the business works:** Customers send a design and get parts or assemblies back in days. Most companies own their plants, which cost a fraction of a precision plant; one leased a second plant and opened it within months. Some route work to partner shops instead, keeping less of each sale: one listed parts maker earns a 49% gross margin in its own plants and 31% on work sent to partners (company filings). Speed is the edge, since one founder says he beats China on price on only about 20-25% of his work (claim). The risk is that customers outgrow them and buy their own machines.

Example companies: SendCutSend, 1872, Senra Systems, Machina Labs, CircuitHub, Jeevy, Protolabs

Precision and defense hardware comes in two factory models. Owned companies run their own plants, keep the whole margin and all the data, and pay the full build before contracts arrive. Hybrid companies mix owned plants with capacity someone else paid for, through franchisees or cells hosted inside a buyer's plant. They keep less of each sale but need far less capital.

Precision & Defense Hardware: owned factories

- **Why it's interesting:** Precision parts are metal or composite parts made to very tight limits, by machining, 3D printing, casting or molding, for missiles, jets, satellites and submarines. The shops that make them are small and slow, and primes, which earn 7-12% on cost-plus contracts, outsource much of the work. A supplier that delivers fast earns a premium and, once approved, keeps the work for programs that can run 50 years.
- **How the business works:** The company sells parts under purchase orders or multi-year contracts. It pays for the plant before approval: one parts maker committed about $200M to a single plant, and a former employee said a six-machine first plant cost "well north of $10 million." Debt and Navy co-funding help once revenue exists. 3D-printing companies keep their printers and sell parts, since selling printers has gone badly. Buying a certified shop from a retiring owner is the cheaper route: the machines and approvals already exist, and roll-ups buy at about seven times annual earnings (claim). New plants need AS9100 certification, the aerospace quality standard, which typically takes 6-12 months, then each part's approval at the buyer.

Example companies: Hadrian, Divergent, Freeform, Amca, Atomic Industries, Fabri, Layup, Hybron, Rangeview, Furientis, Helicon, Westmag, Re:Build, Anduril, Precision Castparts, Howmet

Precision & Defense Hardware: hybrid factories

- **Why it's interesting:** This is the lightest-capital way into precision parts. It answers the capex-before-contracts problem by putting someone else's money into the machines.
- **How the business works:** In a franchise, franchisees buy the machines while the company supplies software, orders and quality rules and keeps a share. One network claims 14 factories, 4 owned and 10 franchised, and a lead-time cut from about 10 weeks to under 14 days for one customer (claims). In a hosted cell, the company runs its machines inside a buyer's plant; one 7-year deal places cells inside a prime's missile plant. The open question is whether primes accept the same quality across many sites.

Example companies: Isembard, Hadrian

C. Specialty Chemicals

We limit chemicals to automated chemistry labs that make new molecules for drug programs. Our view: early, and they need paying customers.

Producers (hybrid)

- **Why it's interesting:** Drug companies send most early chemistry work offshore, largely to China, and the two biggest offshore providers are worth close to $100B combined (Lightspeed). BIOSECURE, a law passed in December 2025, limits use of certain Chinese biotech suppliers, and 79% of 124 biotechs surveyed had a China-based contract manufacturer (BIO). Robotic labs that log every result, including failures, also build data that improves each prediction.
- **How the business works:** The lab sells chemistry work to drug companies, making small batches of new molecules at milligram-to-gram scale for now (claim). It owns the lab, software and data, and does not develop its own drugs. Labs cost tens of millions; one is part-funding an £89.9M expansion with about £22M of UK and Scottish grants. US chemists cost several times Chinese rates, so automation has to bring unit cost down.

Example companies: Chemify, Warp Sciences

D. Advanced Materials

Advanced materials are new materials such as battery cathodes, cooling fluids and alloys. AI can find candidates in weeks, but making them at scale and getting customers to qualify them takes one to five years. Our primer view is selective for producers and early for discovery labs.

Producers

- **Why it's interesting:** China makes about 85% of battery cathode material and over 98% of the cheaper LFP type. Data centers used 415 TWh of power in 2024, heading to about 945 TWh by 2030, and cooling takes 7% to over 30% of it. Both create buyers for a non-Chinese or new material.
- **How the business works:** The producer designs the material and sells it by the kilogram or ton under long qualification and supply deals. Some own the plant; others use a contract maker, a hybrid with far less capex. Owned cathode plants cost hundreds of millions and compete on Chinese prices. Even an established incumbent's battery-materials unit lost money on €436M of revenue.

Example companies: Chemix, Albemarle, Umicore, Corning

Materials Discovery Labs

- **Why it's interesting:** AI plus robotic labs can test far more candidates than people can. One lab made about 1,200 alloys in five or six months (claim), and an independent university test found one of its alloys beat a standard aerospace alloy.
- **How the business works:** Labs earn from paid R&D for partners, licensing, royalties, software licenses and government research awards, while a partner makes and sells the material. They don't own the output, so

their share is a fee or royalty. Government awards run around $1M. Valuations are set on talent, from a $300M seed to one reported at about $7.5B. One investor says materials has had "no AlphaFold moment."

Example companies: Radical AI, CuspAI, Periodic Labs, Lila Sciences, Aionics, Mattiq, Dunia Innovations, Entalpic, Discovered Materials, Kebotix, Citrine Informatics, Uncountable, Albert Invent, Microsoft Discovery

E. Enabling Software & Robotics

These companies sell tools to factories and plants instead of making the product. The primer tracks them outside the core, because selling to plants has been hard and incumbents own the installed base.

Enablers

- **Why it's interesting:** The labor gap and underused plants create demand for tools that raise output. Process plants run at about 74% of capacity in Europe against the 80-85% they need (FTI Consulting), and in a sold-out plant each extra point of yield is close to pure profit. The US installed 38,428 industrial robots in 2025 against 354,226 in China. AI already speeds work around the machine; one tool claims to cut assembly-instruction writing from about 10 hours to 15 minutes (claim).
- **How the business works:** Vendors sell software subscriptions, robots, or robot cells they own and rent out. The closest fit for an enterprise software fund is plant-control software paid on the production it adds (claim): it adjusts the plant's setpoints, such as a temperature or flow rate, itself, with no factory to build. The customer owns the output and the data, big automation vendors bundle software with hardware, and robot-learning success is only 59-66% on short tasks against the 99.9% plus a factory needs.

Example companies: Lumaril, Arda, Mind Robotics, GrayMatter Robotics, CloudNC, AI Robotics, Bright Machines, Rapta, Allus, Faraday, Siemens, Rockwell Automation, Honeywell, AspenTech, Emerson